

CSE-481 ENGINEERING ECONOMICS

INTEGRATED BIST 100 AGENTIC FINANCIAL ANALYTICS HARNESS

Technical Analysis • Sector Relative-Value Research • Fundamental Analysis • Macro Data • News & Speech Intelligence • Backtesting • MCP • Stateful Agent Harness

Educational research system for BIST 100 stocks

Python + Yahoo Finance + Fintables + TCMB/TÜİK/Federal Reserve + News/Speech Evidence + MCP AI Agents

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1. Project Title

Integrated BIST 100 Agentic Financial Analytics Harness: Sector Rotation, Technical Reversals, Fundamental Change, Macro Context, News Intelligence and Backtest Verification

2. Project Overview

This project asks students to build an integrated educational financial-analysis system for a selected universe of BIST 100 stocks. The system does not consist of a single prediction model. Instead, it combines deterministic financial tools, multiple evidence sources, historical verification, and a controlled AI-agent workflow.

Students investigate whether observable market patterns exist before a stock moves, whether lagging stocks within a strong sector later catch up, whether weekday return patterns are stable, whether technical reversal events repeat, and how quarterly financial changes are reflected in subsequent prices. The project also introduces macroeconomic context, curated financial news, and speech-to-text summaries of instructor-approved public finance commentary.

Core Design Principle

LLM reasons and explains. Python tools calculate. The harness controls. Backtests verify. Evidence is logged. A human makes the final decision.

The system is strictly for educational and research purposes. It must not connect to a broker, send real orders, manage real money, or present personalized investment advice.

3. Learning Objectives

- Build reproducible financial data pipelines for BIST 100 stocks and sector-level comparisons.
- Distinguish raw market data, deterministic calculations, model interpretation, and AI-generated explanation.
- Detect sector laggards and test whether relative underperformance is followed by statistically meaningful catch-up behavior.
- Test weekday and multi-day calendar patterns without assuming they are permanent.
- Measure repeated technical reversal behavior using Bollinger Bands, RSI extrema, Supertrend, KAMA and Ichimoku Cloud.
- Link quarterly financial-statement changes to post-disclosure price behavior using point-in-time data.
- Integrate USD/TRY, EUR/TRY, CBRT policy rates, Federal Reserve policy rates and TÜİK inflation indicators.
- Integrate news and public expert commentary as timestamped contextual evidence rather than as unverified financial facts.
- Design an MCP-based agent architecture with constrained tools, state, evidence, memory, verification and human review.
- Understand why backtest performance, prediction accuracy and LLM confidence are different concepts.

4. Financial Problem Definition

The system should answer research questions rather than simply output a BUY or SELL label. A useful output is an evidence-backed analytical state such as 'sector laggard with improving fundamentals', 'historically repeated lower-Bollinger reversal', or 'weekday pattern not statistically stable'.

The four mandatory research scenarios are:

- Sector Relative-Value / Catch-Up Scenario
- Weekday and Multi-Day Pattern Scenario
- Technical Reversal Event Scenario
- Quarterly Fundamental Change and Price-Reaction Scenario

5. Study Universe: 30 BIST 100 Stocks from Multiple Sectors

The following diversified study universe is intended to make sector comparison possible. Sector labels are simplified for coursework. Index membership must be re-checked at the start of each run because BIST index constituents change over time.

Ticker	Yahoo Symbol	Simplified Sector
ASELS	ASELS.IS	Technology / Defense
TUPRS	TUPRS.IS	Petroleum / Refining
BIMAS	BIMAS.IS	Retail Trade
THYAO	THYAO.IS	Transportation / Airlines
AKBNK	AKBNK.IS	Banking
KCHOL	KCHOL.IS	Holding / Investment
EREGL	EREGL.IS	Basic Metals / Steel
TCELL	TCELL.IS	Telecommunications
CCOLA	CCOLA.IS	Food & Beverage
MGROS	MGROS.IS	Retail Trade
SASA	SASA.IS	Chemicals / Plastics
TRALT	TRALT.IS	Mining
FROTO	FROTO.IS	Automotive
ENKAI	ENKAI.IS	Construction
PGSUS	PGSUS.IS	Transportation / Airlines
MPARK	MPARK.IS	Healthcare
EKGYO	EKGYO.IS	Real Estate Investment Trust
GUBRF	GUBRF.IS	Chemicals / Fertilizer
TURSG	TURSG.IS	Insurance
ENJSA	ENJSA.IS	Electric Utilities
TTKOM	TTKOM.IS	Telecommunications
OYAKC	OYAKC.IS	Cement
AKSEN	AKSEN.IS	Electric Utilities
ISMEN	ISMEN.IS	Brokerage / Capital Markets
ARCLK	ARCLK.IS	Durables / Machinery
ECILC	ECILC.IS	Healthcare / Pharmaceuticals
MAVI	MAVI.IS	Retail / Apparel
PETKM	PETKM.IS	Petrochemicals
TOASO	TOASO.IS	Automotive
KRDMD	KRDMD.IS	Basic Metals / Steel

Source note: the selection is based on the BIST 100 constituent universe observed in 2026 and intentionally spans banking, industry, retail, transportation, telecommunications, energy, chemicals, mining, construction, healthcare, insurance, real estate and other sectors.

6. Data Sources

Layer	Example Source	Use in the System
Market Data	Yahoo Finance / yfinance	OHLCV, adjusted prices, BIST index series, sector/peer price histories
Fundamentals	Fintables	Balance sheet, income statement, cash-flow statement, growth, profitability, leverage, liquidity and valuation ratios
Türkiye Macro	TCMB / EVDS	USD/TRY, EUR/TRY, policy-related series, exchange-rate history and monetary indicators
Inflation / Economy	TÜİK	CPI, monthly/annual inflation, industrial production, unemployment and other official statistics
Global Rates	Federal Reserve	Federal funds target range and FOMC policy changes
Financial News	Bloomberg or legally accessible licensed feeds	Timestamped headlines/articles, company and macro event tags, source metadata
Public Expert Commentary	Instructor-approved public YouTube videos	Speech-to-text, topic extraction, stance/context tags and source timestamps

News and video material must be used legally. Students must not bypass paywalls, authentication, licensing restrictions, or platform protections. Public video transcripts should be stored only to the extent needed for educational analysis, with source URL, speaker/channel, publication time and segment timestamps.

7. Technical Analysis

The technical layer should calculate, store and analyze at least the following groups:

Group	Indicators	Research Use
Trend	SMA, EMA, KAMA	Direction, slope, crossovers, distance from adaptive average
Momentum	RSI, MACD	Momentum regime, RSI local extrema, divergence candidates
Volatility	Bollinger Bands, ATR	Band touches, bandwidth, volatility regime
Trend State	Supertrend	Bullish/bearish state changes and post-flip returns
Cloud / Structure	Ichimoku Cloud	Price above/below cloud, cloud breakout/rejection, Tenkan/Kijun relationships
Market Structure	Support/Resistance	Pivot clustering, touches, recency, volume and reversal strength
Activity	Volume / turnover	Confirmation of breakouts and reversal events

8. Technical Reversal Event Framework

Students should treat a technical event as a timestamped research event. The system records the event, the market state at that time, and the forward return after fixed horizons.

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Example event record
Ticker: ASELS
Event: Bollinger Lower Band Touch
Event Date: t
RSI State: Local Low
Supertrend State: Bearish
KAMA Slope: Flattening
Ichimoku State: Below Cloud
Volume Regime: High
Forward Return Horizons: 1d, 3d, 5d, 10d
Reversal Strength: measured after the event
```

Mandatory event families:

- Lower Bollinger Band touch followed by an upward reversal.
- Middle Bollinger Band test followed by continuation or rejection.
- Upper Bollinger Band touch followed by a downward reversal.
- RSI local trough and local peak events; optionally divergence candidates.
- Supertrend direction flips and subsequent returns.
- KAMA slope changes, KAMA-price crosses and adaptive trend persistence.
- Ichimoku cloud breakout, cloud rejection and return into the cloud.
- Support/resistance touches with reversal-strength measurement.

9. Fundamental Analysis with Fintables

Fundamental information must be point-in-time. Historical analysis should use the date on which the financial statement became publicly available, not only the fiscal quarter-end date. This is essential to prevent future information from entering a historical test.

Category	Metrics	Question
Growth	Revenue Growth, EBITDA Growth, Net Profit Growth	Is the company expanding and improving earnings?
Profitability	ROE, ROA, EBITDA Margin, Net Profit Margin	How efficiently is capital converted into profit?

Leverage	Net Debt/EBITDA, Debt/Equity	Is financial leverage improving or deteriorating?
Liquidity	Current Ratio, Quick Ratio	Can short-term obligations be met?
Valuation	P/E, P/B, EV/EBITDA	How is the company priced relative to earnings/equity/operations?
Cash Flow	Operating cash flow and cash conversion	Do accounting profits convert into cash?

10. Feature Engineering

- Daily and weekly returns, rolling volatility, ATR and volume z-scores.
- Relative return versus XU100, sector benchmark and peer median.
- Sector lag score: stock return minus sector-median return over several windows.
- Distance from Bollinger bands, KAMA, support/resistance and Ichimoku cloud boundaries.
- Event-count features: how many similar reversals occurred in the historical window.
- Reversal-strength features over 1, 3, 5 and 10 trading days.
- Weekday and multi-day sequence labels (e.g., Mon-Tue-Wed down, Thu-Fri up).
- Quarterly revenue, EBITDA, net profit, margin, ROE and leverage changes.
- Post-disclosure abnormal return relative to sector and XU100.
- Macro changes: USD/TRY, EUR/TRY, TCMB rate, Fed rate, monthly/annual CPI.
- News event tags and speech-intelligence topic/stance tags with timestamps.

11. Mandatory Research Scenario 1 — Sector Laggard / Catch-Up

Opportunity

Goal: identify stocks that are materially lagging their own sector while the sector is strong, then test whether selected laggards historically tend to catch up before an obvious price surge occurs.

Illustrative Scenario — Banking Sector

Assume AKBNK, GARAN, ISCTR and YKBNK form the peer group. Over the latest 20 trading days, the sector median return is +12%. Three banks gained roughly +11% to +16%, while one bank gained only +3%. The lagging bank is not automatically a BUY. The harness asks whether (a) fundamentals are deteriorating, (b) technical trend is stabilizing, (c) relative weakness has historically mean-reverted, and (d) macro conditions explain the lag. If fundamentals are stable and price begins to recover above KAMA / Supertrend while the sector remains strong, the system may label it 'POTENTIAL_CATCH_UP_CANDIDATE' for backtest verification.

Required analysis:

- Build peer groups by sector.
- Calculate stock return minus sector median/sector index return over several windows.
- Rank laggards and leaders inside each sector.
- Measure whether laggards subsequently outperform peers over 5/10/20 trading days.
- Separate structural laggards from temporary laggards using fundamental and macro evidence.
- Backtest the rule across sectors and periods before presenting any conclusion.
- Report false positives: lagging stocks that continued to underperform.

12. Mandatory Research Scenario 2 — Weekday and Multi-Day Patterns

Goal: test whether individual stocks or sectors exhibit recurring day-of-week or multi-day sequences, while recognizing that apparent patterns may disappear out of sample.

Illustrative Scenario — THYAO

Suppose historical data suggests a sequence in which Monday, Tuesday and Wednesday are more frequently negative, followed by positive Thursday and Friday returns. Students must not assume this is tradable. They should count occurrences, compare average and median returns, test different market regimes, compare against random or unconditional returns, include transaction costs, and evaluate whether the pattern remains in later unseen periods. A second stock may show the opposite pattern.

Examples of patterns to test:

- Monday weakness followed by Friday strength.
- Three consecutive down days followed by a two-day rebound.
- Post-large-down-day mean reversion.
- End-of-week continuation versus reversal.
- Patterns conditional on XU100 direction, sector direction or macro-event weeks.

Students should explicitly discuss multiple-testing risk. Searching hundreds of combinations can create false patterns by chance.

13. Mandatory Research Scenario 3 — Technical Reversal Patterns

Illustrative Scenario — ASELS

The system finds 24 historical lower-Bollinger touches. In 15 cases, price moved higher within the next five trading days. When the lower-band touch occurred together with an RSI local trough, rising volume and a flattening KAMA, the five-day reversal was historically stronger. However, when price remained below a bearish Supertrend and below the Ichimoku cloud, several signals failed. The harness therefore compares single-indicator events against combined-event conditions.

Students should calculate:

- Number of lower-band, middle-band and upper-band touches.
- Bounce/failure count from each band.
- Average/median reversal strength after each event.
- RSI trough/peak behavior and optional divergence candidates.
- Supertrend flip follow-through.
- KAMA-cross and KAMA-slope behavior.
- Ichimoku cloud breakout/rejection statistics.
- Results by market regime: rising XU100, falling XU100, high-volatility and low-volatility periods.

14. Mandatory Research Scenario 4 — Quarterly Fundamentals and Price Reaction

Goal: connect quarterly changes in company financials to price behavior after the information becomes public.

Illustrative Scenario — TUPRS

Assume a quarter shows improving revenue, EBITDA margin and operating cash flow, while Net Debt/EBITDA declines. The system aligns those values with the public disclosure timestamp. It then measures the stock's 1-day, 5-day and 20-day returns after disclosure, and also calculates abnormal return versus both XU100 and a sector peer basket. The same procedure is repeated across many quarters. The research question is not 'good earnings = price rises'; it is whether specific changes were historically associated with different price reactions.

The system should support:

- Quarter-over-quarter and year-over-year changes in revenue, EBITDA and net profit.
- Changes in ROE, margins, liquidity and leverage.
- Valuation changes before and after the reporting period.
- Disclosure-date alignment to avoid look-ahead bias.
- Post-disclosure returns and abnormal returns versus sector/XU100.
- Comparison of positive fundamental surprises with weak technical states, and vice versa.
- Sector-specific metrics where appropriate; bank analysis should not blindly use industrial-company ratios.

15. Macro Context Layer

Variable	Source	Use
USD/TRY and EUR/TRY	TCMB / EVDS	FX-sensitive sectors, exporters, import costs and foreign-currency debt
TCMB policy conditions	TCMB	Domestic discount rate, credit conditions and sector sensitivity
Federal funds target	Federal Reserve	Global risk appetite, dollar funding and international capital conditions
TÜİK CPI	TÜİK	Inflation regime, real returns and company cost/pricing context
Optional macro series	Official sources	Industrial production, unemployment, confidence indicators, commodity prices

Macro variables must be aligned by their publication timestamps. Revised or future data should not be retroactively inserted into a historical decision unless the project explicitly studies revisions.

16. Financial News and Speech Intelligence Layer

The system may enrich structured data with unstructured context from Bloomberg or other legally accessible high-quality financial news feeds and from selected public finance videos.

News pipeline:

- Collect legally accessible headline/article metadata with publication timestamp.
- Tag company, sector and macro topics.
- Extract event categories such as earnings, regulation, FX, rates, commodity shock, contract award or geopolitical risk.
- Optionally calculate a simple sentiment/stance label, but preserve the original source and timestamp.
- Never replace numerical market/fundamental data with a language-model guess.

Speech-to-text pipeline:

- Use instructor-approved public videos from reputable finance educators/practitioners and official financial channels.
- Examples may include public valuation teaching by Aswath Damodaran and public macro/investment commentary from well-known practitioners; the instructor-approved list should be fixed before experiments.
- Convert selected video segments to text, preserving video URL, speaker/channel, publication time and transcript time range.
- Extract topics, claims, stance and uncertainty.
- Numerical claims in commentary must be checked against deterministic data tools before entering the committed evidence set.
- Expert opinion is contextual evidence, not ground truth.

17. Trading Strategy and Finite Action Space

The strategy layer combines technical, sector-relative, fundamental, macro and contextual evidence. It must use a finite action space.

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Allowed educational actions
HOLD
WATCH
INVESTIGATE
POTENTIAL_CATCH_UP_CANDIDATE
TECHNICAL_REVERSAL_CANDIDATE
FUNDAMENTAL_IMPROVEMENT_CANDIDATE
REJECT_SIGNAL
ANALYSIS_UNSAFE
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Optional BUY/SELL labels may be used only as educational backtest labels. No real broker execution is allowed.

18. Backtesting and Verification Harness

Every strategy or discovered pattern should be historically verified before the system presents a strong conclusion.

- Use signal shifting so decisions at time t cannot use the close or information from $t+1$.
- Use financial-statement publication dates, not only quarter-end dates.
- Compare against Buy-and-Hold, XU100 and where possible a sector benchmark.
- Report cumulative return, Sharpe ratio, maximum drawdown, win rate and benchmark difference.
- Include transaction-cost and simple slippage assumptions.
- Use train/validation/test or rolling walk-forward periods for data-mined rules.
- Report the number of signals; a high return from very few events should be flagged.
- Use regime breakdowns and out-of-sample testing for weekday/pattern discoveries.

19. Optional Machine-Learning Prediction

Machine learning is optional and should not replace the main analytical harness. If used, models may predict direction, abnormal return class, or probability of a defined event outcome.

ML metrics and finance metrics must be reported separately.

ML Evaluation	Financial Evaluation
Accuracy	Cumulative Return
Precision	Sharpe Ratio
Recall	Maximum Drawdown
F1	Win Rate
Confusion Matrix	Benchmark Difference

Students must explain why Prediction $\neq$ Profitable Strategy.

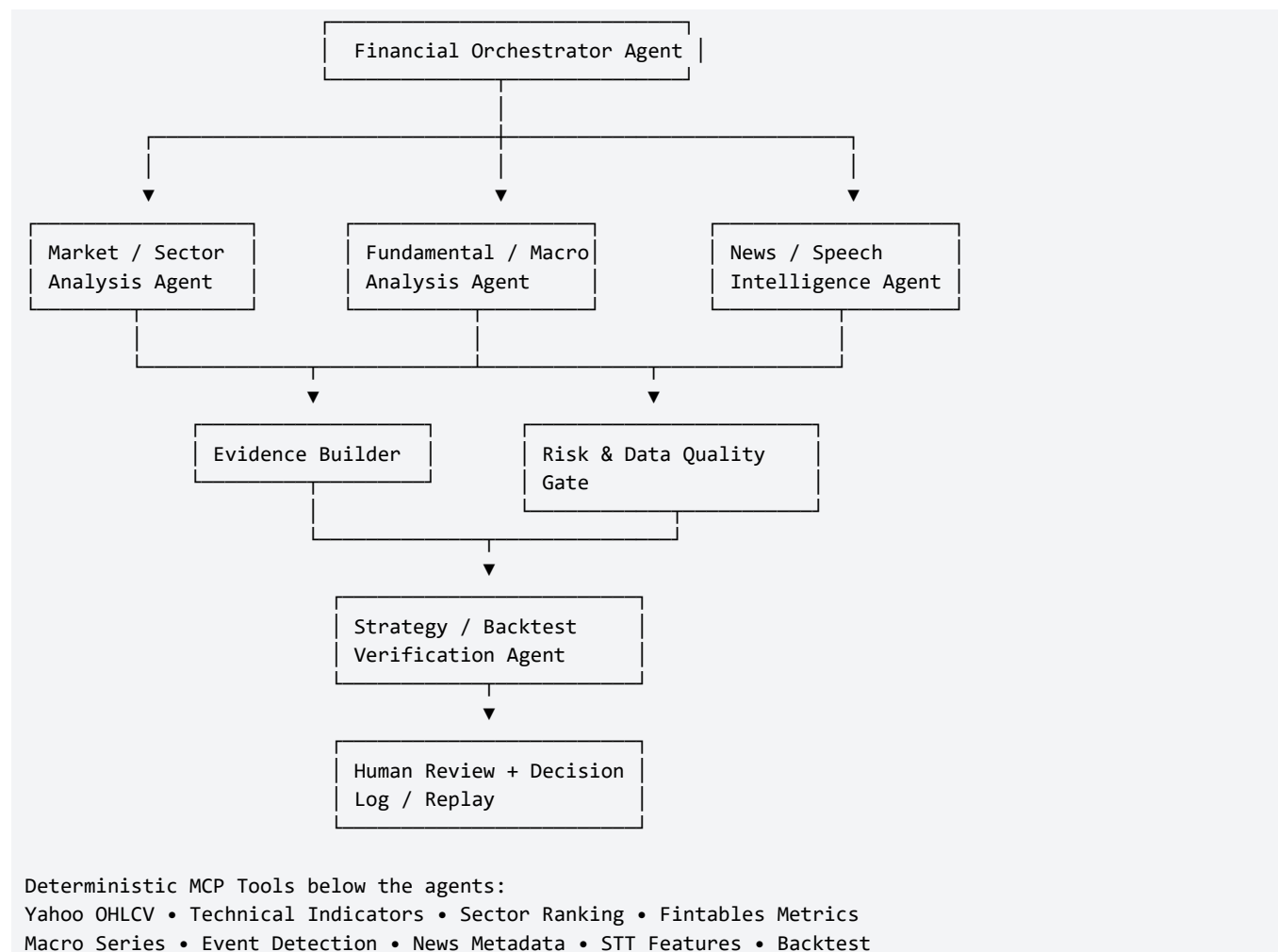
20. Agentic AI Architecture

A small set of specialized agents is recommended. Data collection and indicator calculation remain deterministic services/workers rather than separate reasoning agents.

Agent	Responsibility
Financial Orchestrator Agent	Controls workflow, chooses allowed next state, merges evidence, detects contradictions and requests verification.
Market & Relative-Value Agent	Interprets price/volume, sector lag, weekday patterns, technical indicators and reversal-event statistics.
Fundamental & Macro Agent	Interprets Fintables metrics together with FX, domestic rates, Fed policy and inflation context.
News & Speech Intelligence Agent	Summarizes timestamped news and public commentary while distinguishing opinion, claim and

	verified fact.
Strategy & Backtest Agent	Constructs permitted strategy variants and requests historical verification.
Risk & Data Quality Agent	Checks missing/stale data, period alignment, leakage, look-ahead bias and unsupported conclusions.

21. System Architecture

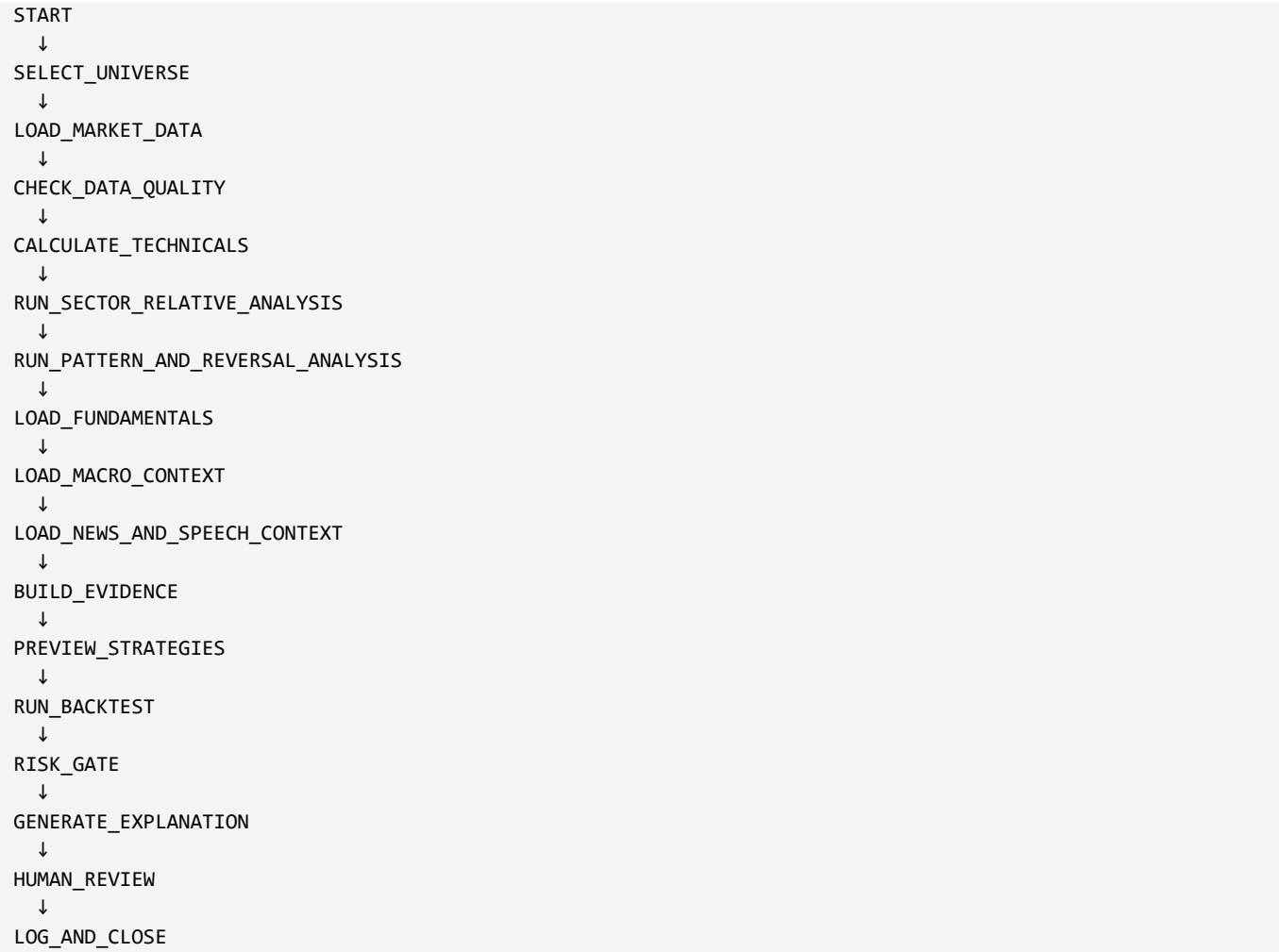


22. MCP Tool Architecture

Tool	Input	Output
get_stock_history	ticker, start, end	OHLCV + metadata
get_index_history	index, start, end	benchmark time series
calculate_indicators	ticker/series, config	SMA/EMA/RSI/MACD/Bollinger/ATR/KAMA/Supertrend/Ichimoku
detect_reversal_events	series, event_config	timestamped technical events
find_support_resistance	OHLCV, tolerance	levels, zones, touch count, strength
rank_sector_relative_strength	sector, horizon	peer returns, lag score, rank
test_weekday_pattern	ticker, pattern, period	occurrences, conditional returns, significance diagnostics
get_fundamental_metrics	ticker, period	point-in-time financial metrics
calculate_fundamental_change	ticker, periods	quarterly/annual changes
get_macro_context	date_range	FX, rates, inflation and publication metadata
get_news_events	ticker/topic, date_range	timestamped legally accessible event metadata
get_speech_features	video/source id	timestamped transcript-derived topics/claims/stance
run_backtest	strategy spec, universe, period	performance + trade/event log
check_data_quality	analysis bundle	warnings, blocks, stale/leakage flags

build_evidence_bundle	analysis id	full and committed evidence sets
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23. Stateful Agent Harness



Each state should expose only a defined subset of MCP tools. The LLM must not invent new actions or skip mandatory quality/backtest states.

24. Memory Architecture

Memory Type	Example
Temporal Memory	Historical prices, indicators, macro series, financial metrics, event histories.
Episodic Memory	Past analysis cases: state → evidence → action → backtest result → human decision.
Procedural Memory	Approved workflow order, tool permissions, validation rules and analysis templates.

A student implementation may use SQLite, PostgreSQL, Parquet/CSV, JSON and Pandas data structures. Enterprise infrastructure is not required.

25. Evidence and Explainability

The system must distinguish between all observed features and the smaller set actually used to justify a conclusion.

Observed Context

- 30+ technical/fundamental/macro/news features

Committed Evidence Set

E1: Stock is 8.5 percentage points behind sector median over 20 days

E2: KAMA slope turned positive

E3: Lower Bollinger reversal historically succeeded 11/17 comparable events

E4: Revenue and EBITDA growth improved in the latest disclosed quarter

E5: Net Debt/EBITDA declined

E6: Macro regime not flagged as hard-risk

E7: Backtest variant B outperformed benchmark after costs

Educational Analytical State:

POTENTIAL_CATCH_UP_CANDIDATE

Warnings:

Historical evidence only; no guarantee of future return.

The AI explanation may cite only evidence that exists in tool outputs.

26. Data Quality and Risk Gate

Hard blocks:

- Future financial-statement information enters an earlier historical date.
- Look-ahead bias in indicator/signal construction.
- Wrong ticker or broken price history.
- Critical period mismatch between price and financial data.
- Backtest uses information unavailable at decision time.

Soft warnings:

- Macro series unavailable for a limited interval.
- News coverage is sparse.
- Video transcript quality is low.
- Small number of historical technical events.
- Sector peer group is temporarily too small.

Possible gate outputs

PASS

PASS_WITH_WARNINGS

INVESTIGATE

ANALYSIS_UNSAFE

HARD_BLOCK

27. Human-in-the-Loop

A human reviewer evaluates the educational analytical conclusion. The system must never silently convert an AI conclusion into a real market order.

Agent conclusion: POTENTIAL_CATCH_UP_CANDIDATE

Human review: REJECT

Reason: historical maximum drawdown is too high

Recorded for: educational decision-memory and later replay

28. Decision Log and Replay

- Ticker and sector
- Analysis timestamp and market-data period
- Data versions and publication timestamps
- Tools called and tool outputs
- Technical / sector / fundamental / macro states
- News and speech source metadata
- Full Evidence Bundle and Committed Evidence Set
- Strategy variant and backtest result
- Risk/Data Quality gate status
- Agent conclusion and explanation
- Human review result and reason

29. Previewable Strategy Space

Before selecting a conclusion, the harness should compare multiple strategy definitions.

Variant	Evidence Used
A — Technical Only	Trend, RSI/MACD, Bollinger, support/resistance, volume
B — Technical + Sector Relative	A + peer lag/leader state
C — Technical + Sector + Fundamental	B + quarterly financial quality/change
D — Full Structured	C + macro context
E — Full Contextual	D + verified news/speech context

Every variant should be previewed through the same backtesting metrics and data-quality rules.

30. Agent Harness Experiments

System Version	Description
A — Raw LLM	All data are passed as text; no deterministic tool requirement.
B — LLM + Tools	MCP tools provide calculations.
C — Tools + Stateful Harness	Tool permissions and ordered states are enforced.
D — C + Evidence + Quality Gate	Committed evidence and blocking rules are added.
E — D + Memory + Human Review	Replayable memory and human decisions are added.

Suggested harness-engineering metrics:

- Numerical hallucination rate
- Invalid tool-call rate
- Unsupported conclusion rate
- Evidence completeness
- Analysis reproducibility
- Backtest correctness
- Number of tool calls
- Latency / workflow cost

31. Evaluation Metrics

Research Layer	Metrics
Sector catch-up research	Subsequent relative return, hit rate, median catch-up size, false-positive rate
Calendar patterns	Occurrence count, conditional mean/median return, out-of-sample stability, cost-adjusted result
Reversal research	Bounce rate, reversal strength, forward return distribution, regime stability

Fundamental reaction	Post-disclosure abnormal return, horizon response, sector-adjusted response
Trading/backtest	Cumulative return, Sharpe, max drawdown, win rate, benchmark difference
ML (optional)	Accuracy, precision, recall, F1, confusion matrix
Agent harness	Hallucination/tool/evidence/reproducibility metrics

32. Suggested Python Functions (Names and Responsibilities)

- load_universe()
- get_stock_history()
- get_index_history()
- build_sector_peer_groups()
- calculate_indicators()
- calculate_bollinger_events()
- calculate_kama_state()
- calculate_supertrend_state()
- calculate_ichimoku_state()
- find_support_resistance()
- calculate_reversal_strength()
- rank_sector_laggards()
- test_weekday_pattern()
- detect_multiday_patterns()
- get_fundamental_metrics()
- calculate_quarterly_changes()
- align_financial_disclosure_dates()
- get_macro_context()
- extract_news_events()
- transcribe_finance_video_segment()
- extract_speech_features()
- build_evidence_bundle()
- generate_strategy_variants()
- run_backtest()
- calculate_performance()
- check_data_quality()
- log_decision()
- replay_analysis()

33. Example End-to-End Harness Scenario

User Question

"Find BIST 100 stocks that are lagging a strong sector but may be entering a historically favorable catch-up state."

1. Orchestrator selects the 30-stock universe.
2. Market tool loads OHLCV and XU100.
3. Data Quality Agent validates dates, missing data and adjustments.
4. Sector tool ranks each stock versus its peers.
5. Technical tools detect Bollinger/KAMA/Supertrend/Ichimoku states and reversal history.
6. Fundamental tool loads only financial data that was public by the analysis date.
7. Macro tool loads FX, CBRT/Fed rate context and TÜİK inflation.
8. News/Speech tool returns timestamped contextual events and verified claims.
9. Evidence Builder separates observed context from committed evidence.

10. Strategy Agent creates several candidate rule sets.
11. Backtest tool evaluates each rule across earlier periods.
12. Risk Gate rejects leakage, weak samples or unstable results.
13. Orchestrator writes an educational analytical conclusion.
14. Human reviewer accepts, modifies or rejects it.
15. Entire workflow is saved for replay.

34. Methodology

1. Freeze the 30-stock universe and sector map for the experiment period.
2. Collect and version market, fundamental and macro datasets.
3. Create point-in-time publication calendars for financial and macro data.
4. Engineer technical, relative-value, fundamental and contextual features.
5. Define event-based research questions before looking at outcomes where possible.
6. Run mandatory sector, calendar, reversal and fundamental-reaction scenarios.
7. Develop several rule-based strategy variants.
8. Backtest with proper signal timing, benchmark comparison, costs and out-of-sample checks.
9. Integrate tools into MCP and implement the stateful harness.
10. Add evidence bundles, memory, quality gates and human review.
11. Run harness experiments A–E and compare reproducibility/safety.
12. Document limitations and cases where the system returns INVESTIGATE or ANALYSIS_UNSAFE.

35. Expected Outcomes

- A reproducible multi-source BIST 100 research dataset.
- A sector-relative ranking system capable of identifying leaders and laggards.
- A tested weekday/multi-day pattern analysis module.
- A technical reversal-event database covering Bollinger, RSI, Supertrend, KAMA, Ichimoku and support/resistance.
- A point-in-time fundamental analysis layer using quarterly Fintables data.
- A macro-context layer using FX, domestic policy context, Fed policy and official inflation data.
- A timestamped news and speech-intelligence context layer.
- A rule-based strategy and verification framework.
- An MCP-based stateful AI agent harness with evidence, memory, tool permissions and replay.
- A clear separation between historical evidence, AI explanation and human judgment.

36. Final Deliverables

- Python source code / notebooks and environment instructions.
- A fixed BIST 100 study universe and sector map.
- Data dictionary describing every structured and unstructured feature.
- Technical reversal-event report.
- Sector laggard/catch-up research report.
- Weekday and multi-day pattern research report.
- Quarterly fundamentals and post-disclosure price-reaction report.
- Macro and contextual evidence integration report.
- MCP tool definitions and stateful harness implementation.
- Decision-log and replay demonstration.
- Backtesting results and benchmark comparisons.
- Agent-harness experiment comparison.
- Final technical report and classroom demonstration.

37. Project Quality Criteria

- Correct point-in-time data alignment and absence of look-ahead bias.
- Clear distinction among tools, workers/services and reasoning agents.
- Reproducible technical and fundamental calculations.
- Meaningful sector-relative, calendar, reversal and quarterly-event experiments.
- Backtests that include benchmarks, costs and risk measures.
- Controlled MCP tool use and explicit harness states.
- Evidence-backed explanations with committed evidence.
- Proper risk/data-quality gates and safe failure states.
- Useful decision logging and replay.
- Critical interpretation of news and expert commentary rather than blind sentiment following.
- Clear limitations and educational-use framing.

38. Final Learning Outcomes

At the end of the project, students should understand that a financial AI system is not valuable because it can produce a confident sentence. It is valuable when its data are correctly timed, its calculations are reproducible, its actions are constrained, its evidence is inspectable, its historical claims are verified, and its conclusions can be challenged by a human.

Key Final Principle

The purpose of the AI agent is not to invent or predict financial facts. Its role is to orchestrate deterministic analytical tools, manage evidence and state, enforce a controlled analysis workflow, and explain reproducible results.

Students should be able to explain the following distinctions:

- Prediction ≠ profitable strategy
- Correlation ≠ causation
- Backtest ≠ future guarantee
- Historical evidence ≠ investment advice
- LLM confidence ≠ statistical probability
- Fundamental strength ≠ guaranteed stock-price increase
- Expert commentary ≠ verified fact

39. Reference Data Sources and Implementation Notes

- Borsa İstanbul — BIST stock indices and periodic constituent reviews (2026).
- BIST 100 constituent/sector reference list observed during 2026; membership should be revalidated for each analysis date.
- Fintables — company financial statements, income statements, cash-flow statements and ratio-analysis pages.
- Central Bank of the Republic of Türkiye (TCMB) — official exchange-rate and monetary/statistical data, including EVDS-linked series.
- Turkish Statistical Institute (TÜİK) — official CPI and economic data portal.
- Federal Reserve Board — FOMC statements and federal funds target range history.
- Yahoo Finance / yfinance — educational retrieval of historical market time series, subject to source availability and terms.
- Bloomberg or other licensed/legally accessible financial news source — contextual event metadata only within permitted access.
- Public YouTube content from instructor-approved financial educators/practitioners — selected segments processed with source and timestamp metadata.

Document note: all numerical examples in the four illustrative scenarios are hypothetical examples designed to explain the required methodology; they are not current market signals or investment recommendations.